Purwadhika Digital Technology School | Digital For Executive

Data Analysis

Session 11: Statistics III

Digital for Executive

Hypothesis Testing

- use case : A/B Testing
- different kind of hypothesis
  - numerical (mean)
    - t test for mean
    - t test 2 independent sample for mean
    - t test paired for mean
    - anova > 2 independent sample for mean
  - categorical (proportion)

	Layout A	Layout B
Visitors	122	118
Customers	22	25
Conversion %	18.0%	21.2%

A/B Testing is hypothesis testing in disguise

Principles of A/B Testing:

- Hypothesis : P.I.C.O.T
- Randomization : bias and covariate
- Sample size : minimal sample size needed
- Method of measurement : categorical data or numerical data

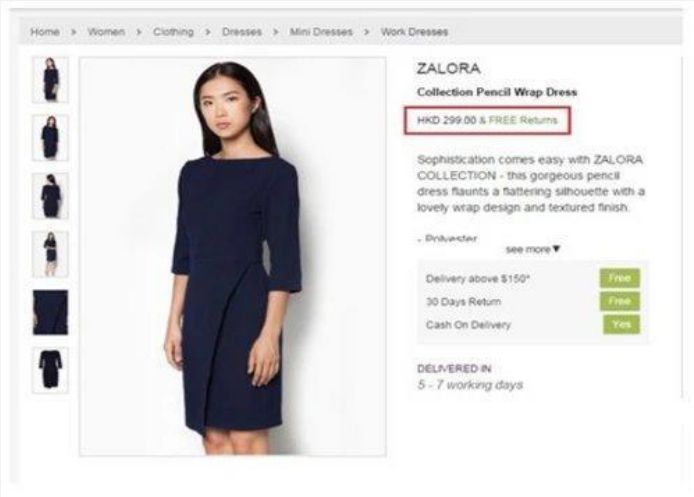
Why do we need A/B testing ?

With A/B we can finally lay to rest the “correlation vs causation” argument , and prove that our new product actually works. As we know that correlation doesn’t imply causation. with A/B testing we proof statistically.

	Layout A	Layout B
Visitors	122	118
Customers	22	25
Conversion %	18.0%	21.2%

“Based on our testing, we infer that **new layout** cause **increase** in **conversion rate**”

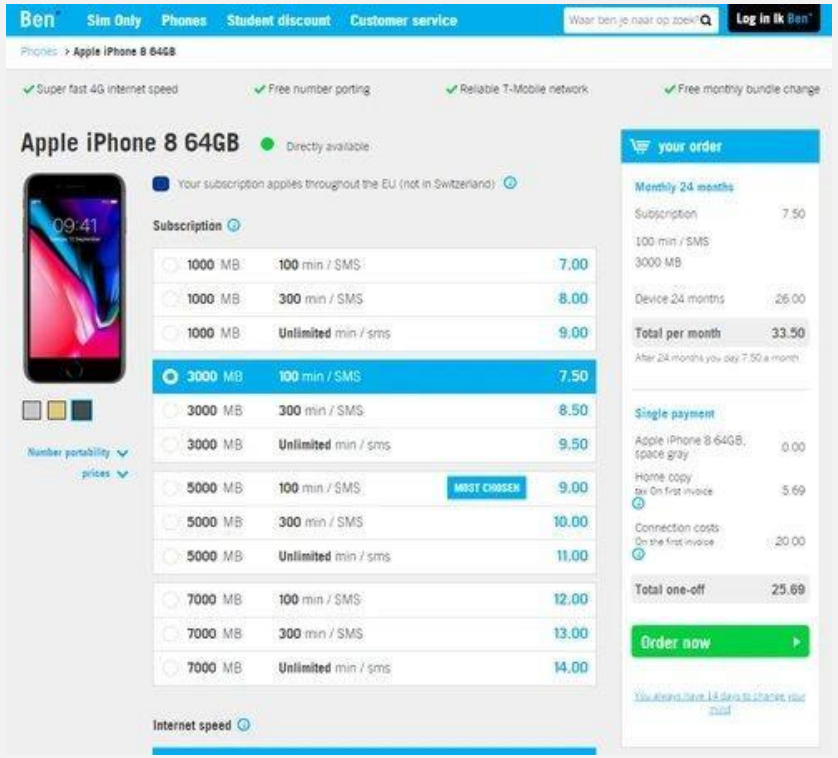
Some Proof in Real Cases : Zalora

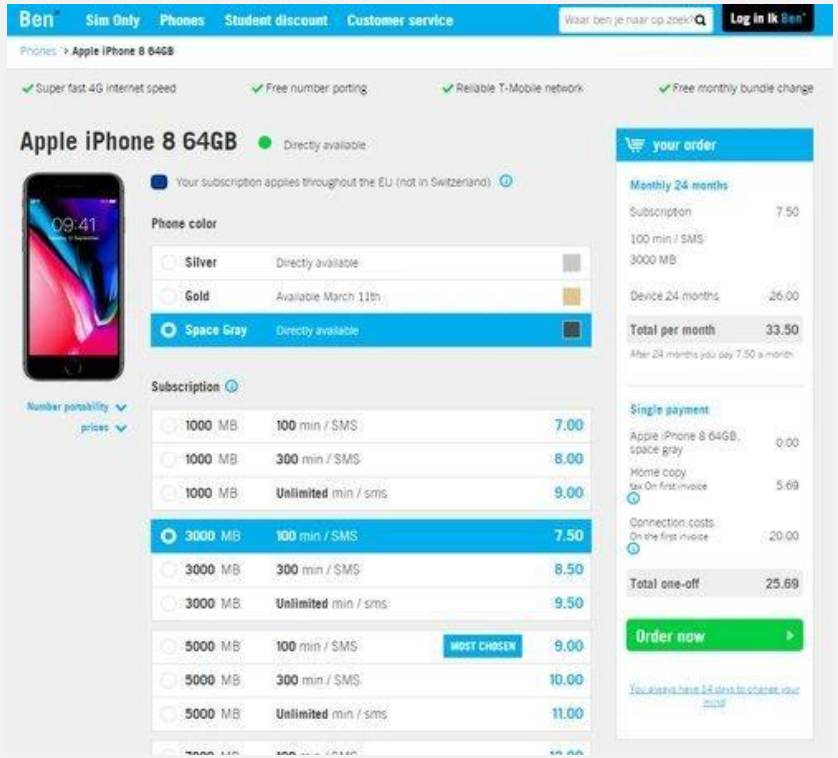


Zalora found that Variation 1 outperformed the control and its counter variation 2. By simply bringing uniformity in Zalora's call to action button, the e-Commerce giant saw an increase of 12.3% in its checkout rate.

Read more at: https://vwo.com/blog/ab-testing-examples/?utm_campaign=tof_ugc

Some Proof in Real Cases : Ben

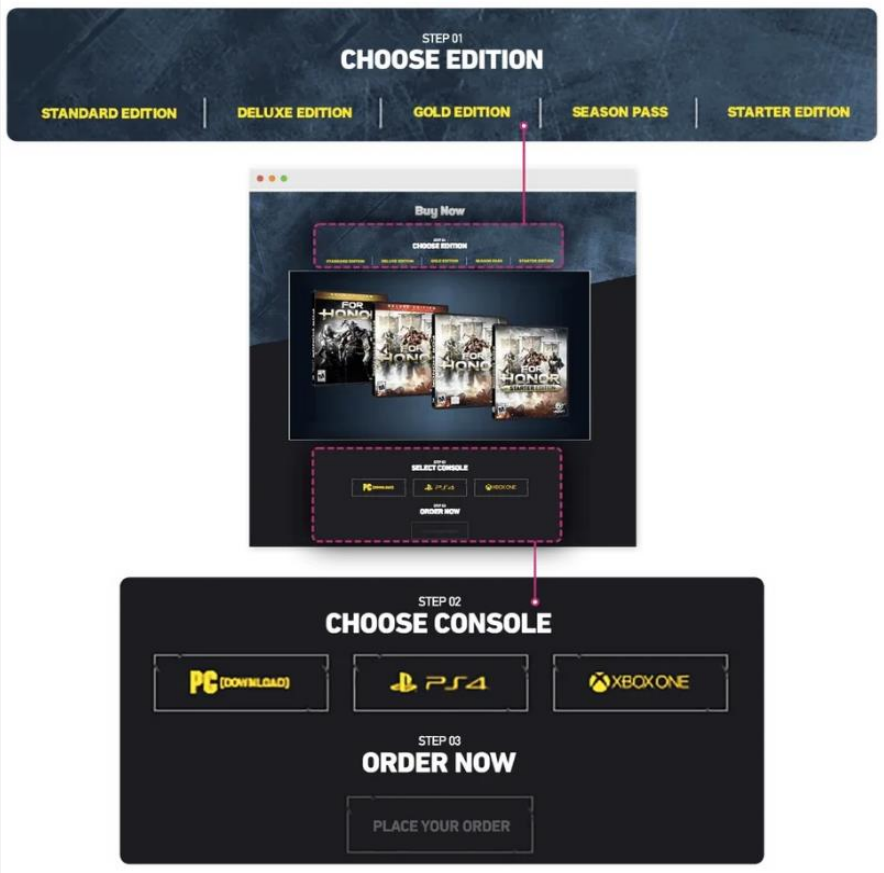




Ben ran the experiment for about two weeks and found that by simply making their color palette more prominently visible, it's conversions went up by 17.63% and the number of customer calls to change device colors dropped significantly.

Read more at: https://vwo.com/blog/ab-testing-examples/?utm_campaign=tof_ugc

Some Proof in Real Cases : Ubisoft



Chose game edition and console (PC, PS4, or X-Box) separately

Some Proof in Real Cases : Ubisoft

Simplify the buying process

Post running the test for about three months, Ubisoft saw that variation brought about more conversions to the company than the control. Conversions went up from 38% to 50% conversions, and overall lead generation increase by 12%.

Hypothesis testing for mean

T-test single population mean

T-test independence double population mean

- equal variance
- non equal variance

T-test paired

Anova F-test for more than two population mean

T-Test Single Population Mean

Assumption and Condition:

1. numerical value with population mean **μo**
2. Data collected using randomization
3. Population distribution approximately normal

Hypothesis:

Ho : **μ** = **μo** (ex. **μ** = 40)

Ha : **μ** ≠ **μo** (two sided) or
μ < **μo** or **μ** > **μo** (one sided)

Ex Age:
39, 40, 41, 44, 45, 38, 42, 43, 45, 46, … 48
Ho : **μ** = 40)

Test Statistics *t* with degree of freedom n - 1:

$$t = \frac{(\bar{x} - \mu_0)}{se} = \frac{(\bar{x} - \mu_0)}{s/\sqrt{n}}$$

Rejection Criteria: P-value ≤ α

Use Case : A/B Testing numerical measurements

A financial tech company want to test whether their new app design help people to increase the amount of money that they save (saving rate). Savings rate is the percentage of each user's monthly paycheck that he or she saves

1. Six months ago, our client randomly selected 1,000 newly signed up users and assigned randomly 500 of them to the current design and 500 to the new design.
2. The control group went on to use the current app.
3. All users started with a 0% savings rate.
4. The 1,000 users represent just a small portion of the app's total users.

T-Test Independence Double Population Mean

Assumption and Condition:

1. numerical variable in each group
2. Data collected using randomization
3. Population distribution approximately normal for each group

Hypothesis:

Ho : **μ1 = μ2** (ex. **μ1 - μ2** = 0)

Ha : **μ1 ≠ μ2** (two sided) or
μ1 < μ2 or **μ1 > μ2** (one sided)

Test Statistics *t*:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{se} \text{ where } se = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Rejection Criteria:

- P-value ≤ α
- in spicy, if you do one-sided hypothesis testing the p-value needed to be divided by two

T-Test Independence Double Population Mean Equal Variance

Assumption:

1. numerical variable in each group
2. Data collected using randomization
3. Population distribution approximately normal for each group
4. Equal variance in each group

Hypothesis:

Ho : **μ1** = **μ2** (ex. **μ1** - **μ2** = 0)

Ha : **μ1** ≠ **μ2** (two sided) or

μ1 < **μ2** or **μ1** > **μ2** (one sided)

Test Statistics *t*:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{se}$$

$$se = \sqrt{\frac{s^2}{n_1} + \frac{s^2}{n_2}} = s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$s = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Rejection Criteria (spicy) :

P-value ≤ α (two-sided)

P-value/2 ≤ α (one-sided)

T-Test Paired

Assumption:

1. numerical variable in each group
2. Data collected using randomization
3. Dependent Sample (each sample has matched observation/repeated measurements)
4. Population distribution approximately normal for each group

Hypothesis:
Ho : **D** = 0

Ha : **D** ≠ **0** (two sided) or
D < **0** or **D** > **0** (one sided)
D is mean of difference population between matched pair

Example, weight before and after 2 months

Before	60	65	60	57	...	67
After	60	63	59	58	...	67
After-before	0	2	1	-1	...	0

Data we actually use to test

Test Statistics *t* with df n - 1:

$$t = \frac{\bar{x}_d - 0}{se} \text{ with } se = s_d/\sqrt{n}$$

Rejection Criteria (spicy) :

P-value ≤ α (two-sided)

P-value/2 ≤ α (one-sided)

Use Case : A/B Testing numerical measurements more than two groups.

A financial tech company want to test whether their new two app design help people to increase the amount of money that they save (saving rate). Savings rate is the percentage of each user's monthly paycheck that he or she saves

1. Six months ago, our client randomly selected 1,500 newly signed up users and assigned 500 of them to the current design and each 500 to the new design (design 1 and design 2).
2. The control group went on to use the current app.
3. All users started with a 0% savings rate.
4. The 1,500 users represent just a small portion of the app's total users.

Anova F-Test for More Than Two Population Mean

Assumption and Condition:

1. The population distribution of the numerical variable is assumed to be normal
2. Data collected using randomization
3. Each group has equal variance

Hypothesis:

Ho : **μ1** = **μ2** = … = **μk** (there are k groups)

Ha : at least one pair of the population are not equal (and we don't know which)

Test Statistics *F* with df1 k - 1 and df2 n - k:

$$F = \frac{\text{Between-groups variability}}{\text{Within-groups variability}}$$

Rejection Criteria (spicy) :

P-value ≤ α

Hypothesis test for proportion

Z test single population proportion

Z test double population proportion

Chi-Squared test more than two population proportion

Use Case : A/B Testing categorical measurements

1. An newly launched online store want to test whether their conversion rate is greater than 50% or not. So they **gathered 1000** visitor **data randomly** and observed whether they are buying or not. As a Data Scientist you are asked to perform a hypothesis testing whether their conversion rate is greater than 50%?

1. A few months later the online store launched a new web design. They **randomly assign** visitor to layout A or layout B and 1000 for each design. You are asked again to perform hypothesis testing whether the conversion rate is increasing or not ?

Similar Ex. Amazon, Ben, Ubisoft

Z-Test Single Population Proportion

Assumption and Condition:

1. Dichotomy categorical variable with population proportion ***po***
2. Data collected using randomization
3. sample size (n) large enough:

n x ***po*** ≥ 15 and n x (1 - ***po***) ≥ 15

Hypothesis:

Ho : ***p*** = ***po*** (ex. ***p*** = 0.5)

Ha : ***p*** ≠ ***po*** (two sided) or
p < ***po*** or ***p*** > ***po*** (one sided)

Test Statistics z:

$$z = \frac{\hat{p} - p_0}{se_0} = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$

Rejection Criteria (spicy) :

P-value ≤ α (two-sided)

P-value/2 ≤ α (one-sided)

Z-Test Double Population Proportion

Assumption:

1. Dichotomy categorical variable observed in two group
2. Data collected using randomization
3. sample size (n1 and n2) large enough that there are five successive and five failure in each group.

Hypothesis:

Ho : ***p1*** = ***p2*** (ex.***p1*** - ***p2*** = 0)

Ha : ***p*** ≠ ***po*** (two sided) or
p1 < ***p2*** or ***p1*** > ***p2*** (one sided)

Test Statistics z:

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{se_0} \text{ with } se_0 = \sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

where $\hat{p}$ is the pooled estimate.

Rejection Criteria (spicy) :

P-value ≤ α (two-sided)

P-value/2 ≤ α (one-sided)

Use Case : A/B Testing categorical measurements more than two groups

A few months later the online store launched a new web design. They **randomly assign** visitor to layout A (current design), layout B1, or layout B2 and 1000 for each design. You are asked again to perform hypothesis testing whether the conversion rate is increasing or not and which design perform best?

Similar Ex. Zalora Case

Chi-Squared Test for Independence between Two Categorical Variables

Assumption:

1. Two categorical variable :

   check out (yes or no) vs web design (design A, B1, B2)

1. Randomization

Hypothesis:

Ho : The two variables are independent

Ha : The two variables are dependent.

Test Statistics Chi-Squared:

$$X^2 = \sum \frac{(observed\ count - expected\ count)^2}{expected\ count}$$

Rejection Criteria: P-value ≤ α

Goodness of Fit Test

- We need normality assumption for some test
- We can check the distribution but that is very vulnerable to subjectivity
- To Test if sample data follow certain distribution (ex. Normal)
- Commonly used goodness of fit test:
  - chi - square
  - anderson - darling
  - shapiro - wilk
  - kolmogorov - smirnov

Hypothesis

Ho : The sample follow certain distribution

Ha : The sample do not follow the distribution

Rejection Region : P-Value < α

Non-Parametric

Non-parametric Statistics usually used when

- we don't have enough data (small sample size)
- we can't assume that our data population follow certain distribution. ex. result of our normality test is reject Ho

	Parametric	Non-parametric
Assumed distribution	Normal	Any
Assumed variance	Homogeneous	Any
Data set relationships	Independent	Any
Usual central measure	Mean	Median
Benefits	Can draw more conclusions	Simplicity; Less affected by outliers
Tests		
Correlation test	Pearson	Spearman
Independent measures, 2 groups	Independent-measures t-test	Mann-Whitney test
Independent measures, >2 groups	One-way, independent-measures ANOVA	Kruskal-Wallis test
Repeated measures, 2 conditions	Matched-pair t-test	Wilcoxon test
Repeated measures, >2 conditions	One-way, repeated measures ANOVA	Friedman's test

Hypothesis testing for mean : Non-Parametrics

- **Mann-Whitney** independence double population mean
- **Wilcoxon** paired mean test
- **Kruskal-Wallis** for more than two population mean

These method are usually used when normality assumption doesn't meet:

- For example : income data or money related data usually have right skew distribution

Wilcoxon

Assumption:

1. numerical value with population mean **Mo**
2. Data collected using randomization
3. Test for median of population
4. sample size too small

Hypothesis:

Ho : **M** = **Mo** (ex. **μ** = 40)

Ha : **M** ≠ **Mo** (two sided) or

M < **Mo** or **μ** > **μo** (one sided)

Test Statistics : Sumrank

Rejection Criteria (SciPy) :

P-value ≤ α (two-sided)

P-value/2 ≤ α (one-sided)

Mann Whitney

Assumption:

1. numerical variable (ordinal, interval, ratio) in each group
2. Data collected using randomization
3. Test for median of population
4. sample size too small

Hypothesis:

Ho : **M1** = **M2** (ex. **M1** - **M2** = 0)

Ha : **M1** ≠ **M2** (two sided) or

M1 < **M2** or **M1** > **M2** (one sided)

Test Statistics : U-Statistics

Rejection Criteria (SciPy) :

P-value ≤ α (two-sided)

P-value/2 ≤ α (one-sided)

Kruskal-Wallis

Assumption:

1. The population distribution of the numerical variable is assumed to be normal
2. Data collected using randomization
3. Test for median of population
4. sample size too small

Hypothesis:

Ho : **M1** = **M2** = … = **Mk** (there are k groups)

Ha : at least one pair of the population are not equal (and we don't know which)

Test Statistics : V-Statistics

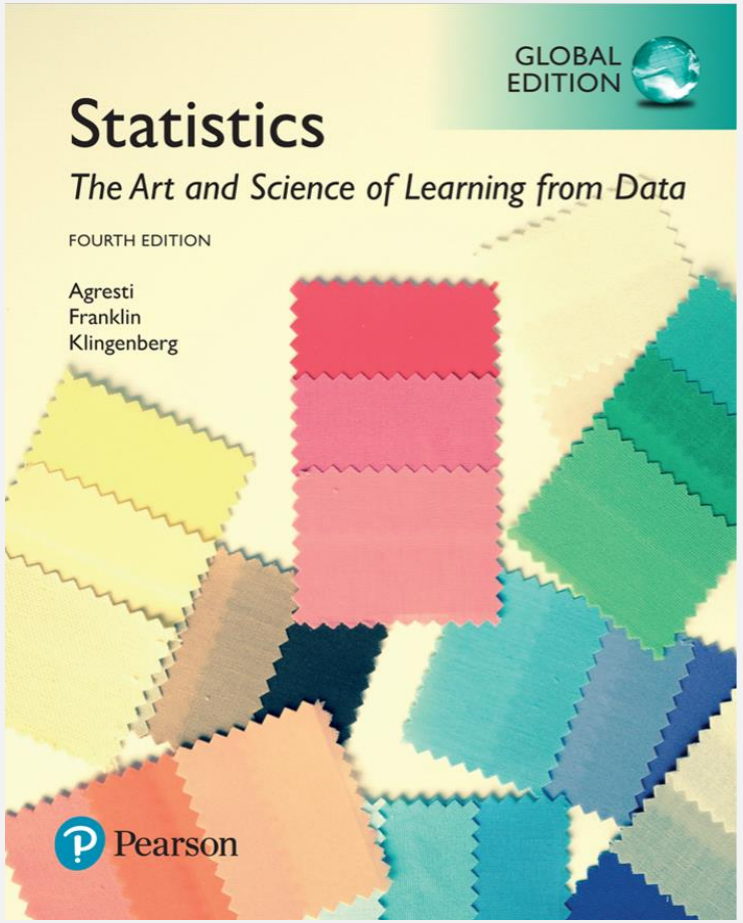
Rejection Region : p-value < α

Another example of Hypothesis testing

Methods that you have learned can also be applied to another cases:

- Medicine, to understand if a drug works or not
- Economics, to understand human behaviour
- Foreign aid and charitable work (the reputable ones at least), to understand which interventions are most effective at alleviating problems (health, poverty, etc)

Reference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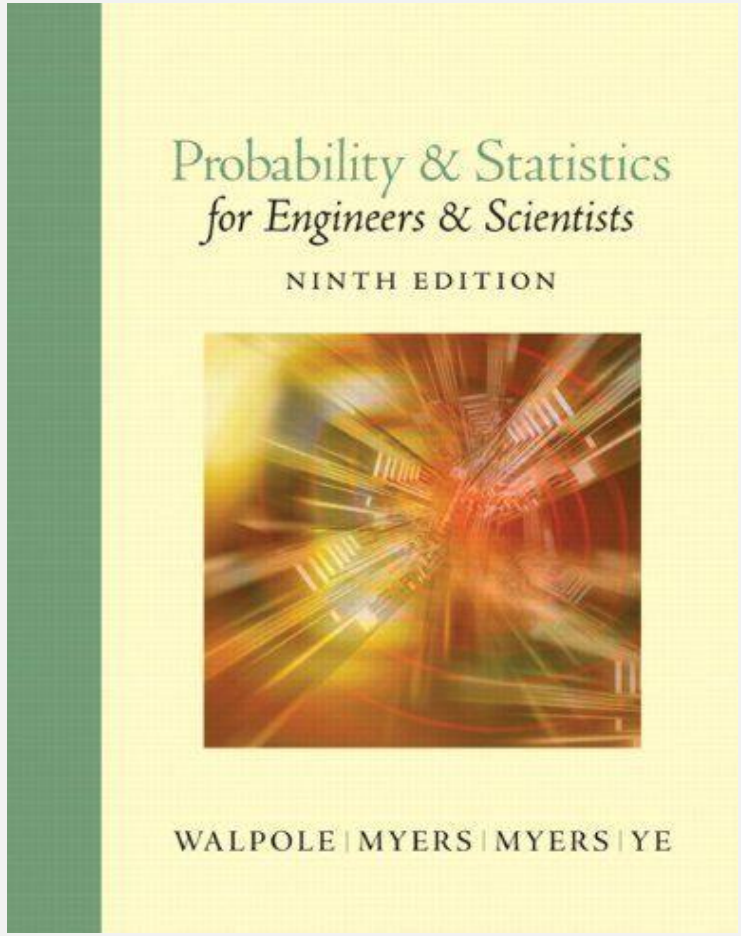
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Thank You!

PURWADHIKA HEAD OFFICE

BSD Green Office Park 9 - G Floor,
BSD City, Tangerang, Banten

CAMPUS BRANCHES

Purwadhika BSD
Purwadhika Jakarta
Purwadhika Bandung

CONTACT

021 - 5012 3161
0813 - 1867 8671
0812 - 8060 7160

EMAIL

admission1@purwadhika.com
admission2@purwadhika.com
admission3@purwadhika.com

WEBSITE

www.purwadhika.com